

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Mining Engineering Department

Commented [U1]: Please enter the name in Arial 12 font

2. **Subject Name:** Material Handling and Mineral Engineering

3. **L-T-P:** 3-0-0 **Credits:** 3

Commented [U2]: Please enter the details (use Arial 12 font)

4. **Pre-requisites:** Nil

Commented [U3]: Please enter the pre-requisites (subject name and subject no. in Arial 12 font)

5. **Syllabus and reference books:**

Commented [U4]: 1. Please type the syllabus and reference books in the box provided below.

Syllabus

- Importance of Mineral Processing
- Particle Characterization
- Crushing and Grinding
- Screening
- Classification
- Gravity Concentration
- Magnetic & Electrostatic Separation
- Mass Balancing
- Fundamentals of Sampling
- Material Storage
- Conveying Systems
- Slurry Transportation
- Flowsheet Development

Tutorials on Grade - Recovery, Mass Balancing, Particle Size Distribution, Circulating Load Estimation, Tromp Curve, Screening Efficiency, Conveying Systems, Critical Velocity and Pumping Systems

Reference Books:

- 1) Wills Barry A., Finch James (2015) An Introduction to the Practical Aspects of Ore Treatment and Mineral Recovery, Wills' Mineral Processing Technology, 8th Edition.
- 2) Fruchtbach Jacob (1988) Bulk Materials Handling Handbook, Springer.
- 3) Rivkin Michael (2018) Bulk Material Handling: Practical Guidance for Mechanical Engineers, Partridge Publishing Singapore.

Please use "Arial" font with a font size of "12" and a line spacing of 1.0 for syllabus and reference books. Please use Justified Text (Ctrl+J) for the content entered in the boxes
Reference books should be numbered as 1), 2), 3)
Please provide all bibliographic details for the reference books

6. **Lecture-wise break-up:**

Commented [U5]: Please provide lecture-wise break-up in the table given below:

Please use "Arial" font with a font size of "12" and a line spacing of 1.0

Commented [U6]:

Sl. No.	Topic	No. of lectures
1.	Importance of Mineral Processing	1
2.	Particle Characterization	1
3.	Crushing and Grinding	3
4.	Screening	1

5.	Classification	1
6.	Gravity Concentration	2
7.	Magnetic & Electrostatic Separation	2
8.	Mass Balancing	2
9.	Fundamentals of Sampling	2
10.	Material Storage	3
11.	Conveying Systems	6
12.	Slurry Transportation	2
13.	Flowsheet Development	2
14.	Tutorials on Grade - Recovery, Mass Balancing, Particle Size Distribution, Circulating Load Estimation, Tromp Curve, Screening Efficiency, Conveying Systems, Critical Velocity and Pumping Systems	14
Total number of hours		42

Commented [U7]: Please convert the template into PDF and upload it in ERP